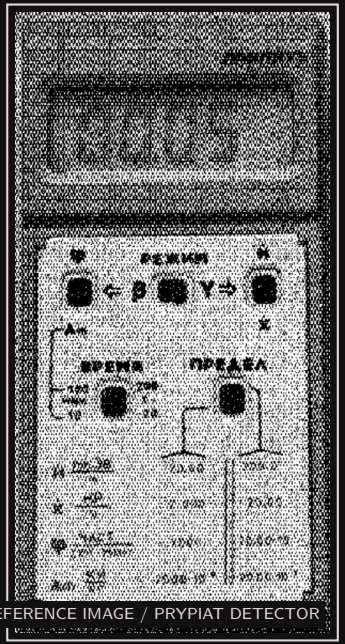
LM-IDNet

Can four traffic counts reveal abnormal IoT behaviour?

4 counts/device ▪ 8/9 attack intervals
Explainable Dirichlet–multinomial detection

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Supervisor: Professor Mauro Migliardi
Co-supervisor: Professor Alessandro Languasco
Discussion presentation ▪ 20 October 2026



REFERENCE IMAGE / PRYPIAT DETECTOR

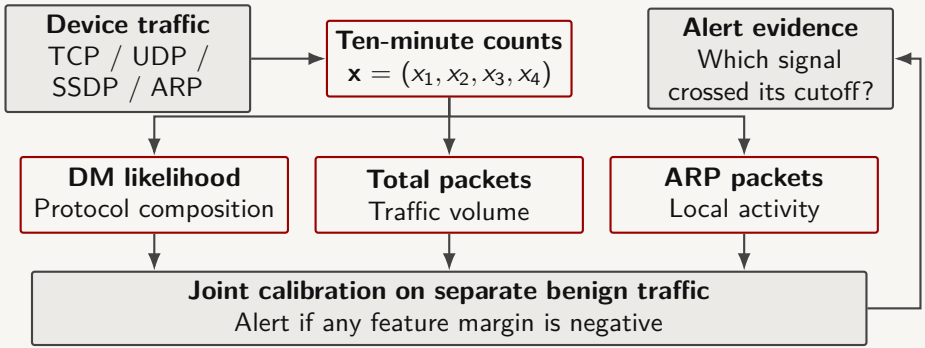
What exists—and where this thesis fits

Research systems	Commercial systems	LM-IDNet
N-BaIoT, Kitsune and DÏoT use richer neural, ensemble or federated representations.	Agentless discovery, broad telemetry, threat intelligence and SOC response.	Four explicit counts, a device-specific normal model and inspectable decision margins.
Strong modelling capacity, but different data and protocols prevent a direct metric ranking.	Microsoft Defender for IoT and Palo Alto Device Security address a wider operational scope.	Reproducible computation, guarded state changes and constrained-runtime evidence.

The added value is compactness and inspectability—not a claim to replace a commercial platform.

Architectural comparison only. The cited systems use different features, datasets and evaluation protocols.

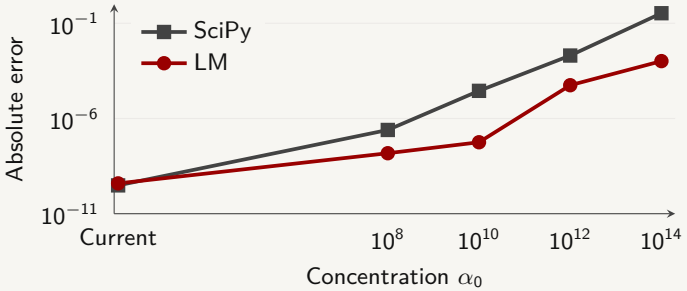
Four counts, three signals, one explainable decision



DM models extra-multinomial variation ($\phi = 218.96$ / 86.84); LM evaluates its likelihood.

DM is the probability model; LM is its numerical backend. Later DM/IoT work: Al-Haj Baddar, Languasco and Migliardi.

Where LM adds numerical value



0 / 1300

decision differences on tested development windows

At 10^{10}: $\sim$**500**$\times$ **lower**

At 10^{14}: $\sim$**327**$\times$ **lower**

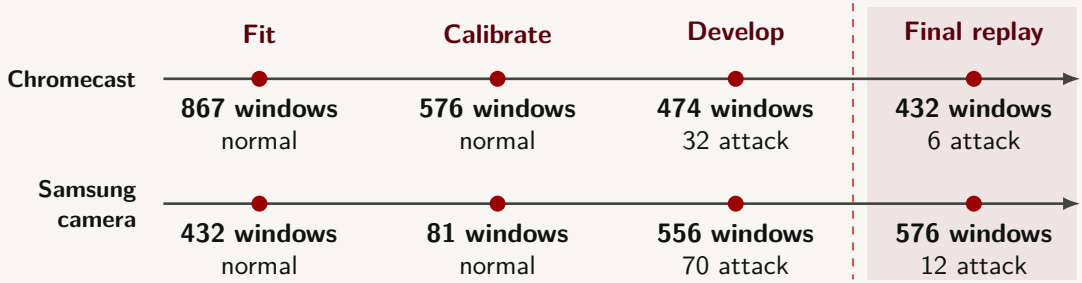
absolute error with LM

Live score: LM 12.04 μs vs SciPy 13.39 μs (10.1% faster). **Initial/periodic fitting:** SciPy is faster.

Current data are easy for both; LM protects accuracy near the multinomial boundary.

100-digit reference probe. Runtime is workload-specific; ten-minute collection dominates alert delay.

Datasets and chronological partitions

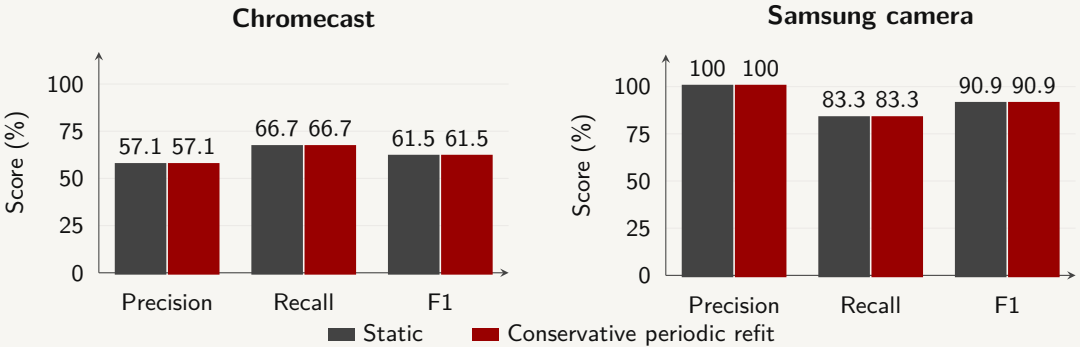


Useful but imperfect test bed: $\psi = 0.0356$ for Chromecast and 0.0575 for Samsung.

Only Chromecast passes the papers' empirical $\psi < 0.05$ convention; both training sets show extra-multinomial variation. Labels are used only for metrics.

Final detection results by device

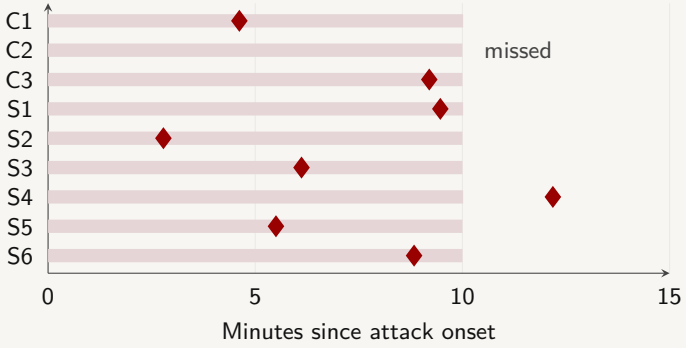
Ten-minute windows ▪ Final partitions



Chromecast: 4/6 attack windows flagged. Samsung camera: 10/12 flagged.

Periodic-refit results match static results for each device; no candidate update was activated.

Attack coverage and detection delay



First alert
Annotated attack

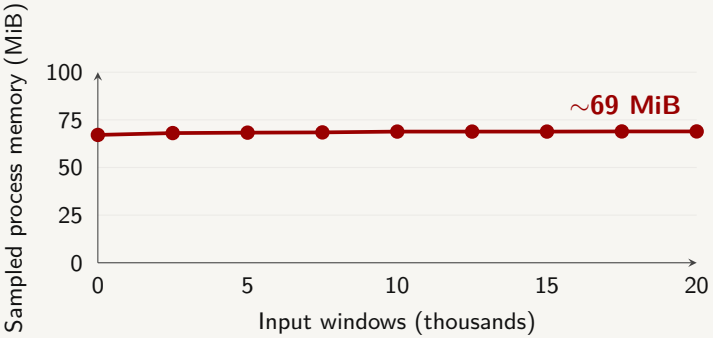
Chromecast
2/3 detected; 2/3 by end
6.9 min median

Samsung camera
6/6 detected; 5/6 by end
7.5 min median

The ten-minute collection interval is part of the detection delay.

Static mode, final official intervals; C = Chromecast, S = Samsung. Median delay is among detected attacks; alerts occur at window close.

What this thesis adds—and what remains



4 counts
compact, interpretable device state

8 / 9
attack intervals alerted; 7 by attack end

~69 MiB
recorded runtime footprint

Limits: two labelled devices and 18 final attack windows; count-preserving attacks can be missed; runtime is simulated x86_64, not a physical gateway; useful adaptation remains unproven.

A compact, reproducible detector whose decisions, numerical core and state changes are inspectable.

20,000 synthetic windows; x86_64 Docker, 1 CPU / 512 MiB; 5.079 s maximum with a rejected refit.

Thank you

Questions and discussion

LM-IDNet ▪ Fatemeh Mahvari

Reference slides follow: model, profiles, native method, counts, guards, window study, sources.

R1 / Model and decision rule

DM composition model $\alpha_k > 0$, $\alpha_0 = \sum_k \alpha_k$, $N = \sum_k x_k$

$$\ell(\mathbf{x}) = \log \frac{N!}{\prod_k x_k!} + \log \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + N)} + \sum_k \log \frac{\Gamma(\alpha_k + x_k)}{\Gamma(\alpha_k)}.$$

Three signals $\ell(\mathbf{x})$, $-\log(1 + N)$, $-\log(1 + x_{\mathrm{ARP}})$.

Fit-only medians and IQRs scale each signal. The minimum scaled signal is calibrated jointly at the 1% lower quantile on separate benign windows.

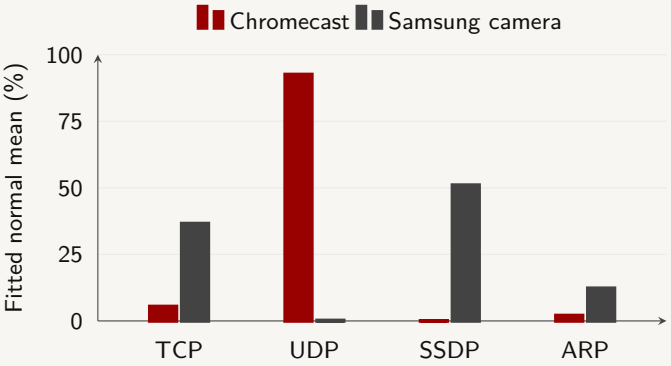
The event score is the minimum signed margin from the resulting cutoffs:

$$s = \min(m_{\mathrm{DM}}, m_N, m_{\mathrm{ARP}}), \qquad \text{alert if } s < 0, \qquad \text{severity} = \max(0, -s).$$

During updates, original DM and count references remain active detection checks.

Constant fit features use unit scale. A count-tail breach can decide early. Severity is not an attack probability.

R2 / Device traffic profiles



Device-specific models

Normal protocol proportions differ between devices.

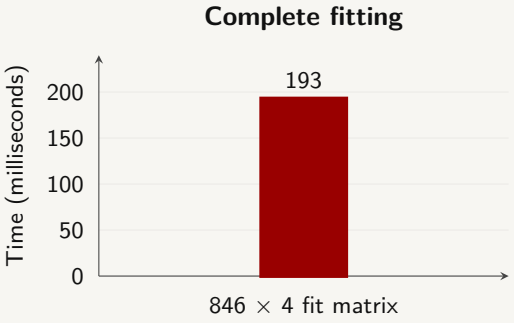
4 exclusive counts
per ten-minute window

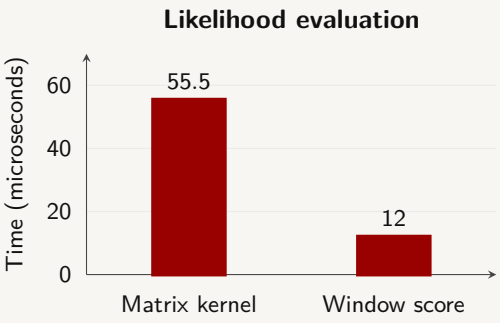
Learn from a trusted benign period; flag unusual later behaviour.

A separate normal model captures each device's traffic pattern.

Fitted means from saved normal-only models. UDP excludes SSDP; bars are model expectations.

R3 / Native Languasco–Migliardi implementation

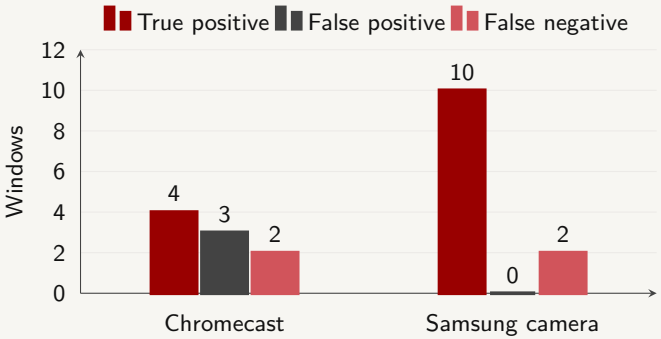




Native integration preserves the sum of independent window likelihoods.

Retained D-Link native-method timings, five batches and warm-up. Fitting excludes input/output. These are recorded measurements.

R4 / Final confusion counts by device



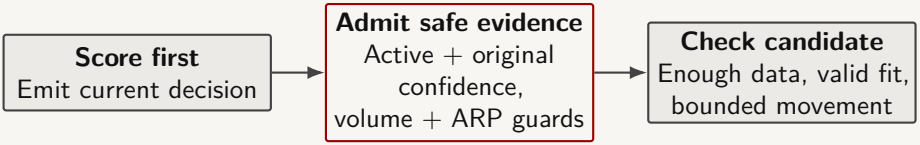
Chromecast
432 windows; 6 attack
TP = 4 FP = 3
FN = 2 TN = 423
FPR = 3/426 = 0.70%

Samsung camera
576 windows; 12 attack
TP = 10 FP = 0
FN = 2 TN = 564
FPR = 0/564 = 0%

Window recall and attack-interval coverage answer different questions.

Static mode, final partitions. A window is attack-labelled if it overlaps an official attack interval.

R5 / Conservative learning guards



$$1/1.25 \leq \alpha_0'/\alpha_{0,t} \leq 1.25 \qquad \text{concentration ratio}$$

$$\max_k |p_k' - p_{k,t}| \leq 0.05 \qquad \text{protocol proportions}$$

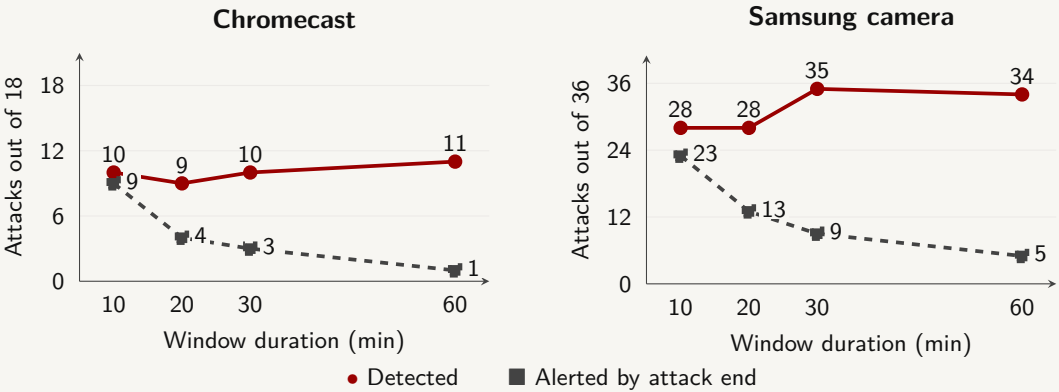
$$|\tau' - \tau_t| \leq 2\,\mathrm{IQR}_0 \qquad \text{threshold movement}$$

Defaults: 288-window buffer; attempt every 144 scored windows; 144 accepted windows required for periodic fitting.

Rejecting a candidate retains the complete active pair; accepted pairs support rollback.

Per-update guards do not establish immunity to stealthy attacks or unlimited cumulative drift. Final replay promoted no candidates.

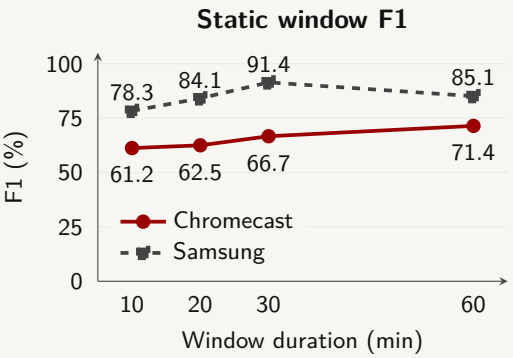
R6 / Window duration: attack coverage

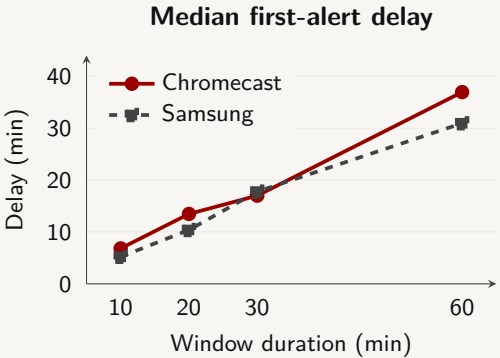


Delivered reference: ten minutes. Longer windows increase response delay.

Static development study; matched traffic within each device; fresh fitting and calibration at every duration.

R7 / Window duration: F1 and detection delay





F1 changes with label granularity; median delay is conditional on detection.

Static development study. Samsung uses May 28–29 to fit and May 30–31 to calibrate at every duration. Lines join measured settings.

R8 / References and project evidence

Statistical learning

Minka, *Estimating a Dirichlet distribution*, 2000 / revised 2012.
Fixed-point estimation of the DM parameters.

Native likelihood method

Languasco and Migliardi, *Computational Statistics* 38, 2023.
doi: 10.1007/s00180-022-01311-7
Al-Haj Baddar, Languasco and Migliardi, *IEEE TPAMI* 47(2), 2025.
doi: 10.1109/TPAMI.2024.3489645

Research and market context

N-BaIoT, Kitsune and DÏoT: neural, ensemble and federated research systems.
Microsoft Defender for IoT and Palo Alto Device Security: representative commercial scope.

Project evidence

Final protocol, results and events: `reports/final_evaluation/`
Duration study: `reports/window_size/`
Backend benchmark: `reports/D-LinkDayCam5/`
Numerical probe: `docs/lm_professor_architecture_experiment_report.md`
Supplied-window simulation: `reports/edge/`